INCH-POUND

MIL-R-82879
20 December 1993

MILITARY SPECIFICATION

REMOVER, CARTRIDGE ACTUATED, RAU-3/A ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the RAU-3/A Cartridge Actuated Remover Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification For
MIL-C-5541	Chemical Films and Chemical Film Material for Aluminum and Aluminum Alloys
MIL-R-82689	Release, Firing Pin, Gas Actuated, M1A1

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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MIL-STD-413	Visual Inspection Guide for Elastomeric O-Rings
MIL-STD-453	Inspection, Radiographic
MIL-STD-1168	Ammunition Lot Numbering
MS28741-4	Hose Assembly, Detachable End Fitting, Medium Pressure

(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Indian Head Division, Naval Surface Warfare Center (CAGE Codes 19200, 19203, and 53711)

DL5184966	M1A1 Firing Pin Release
DL6783541	Remover, Cartridge Actuated, RAU-3/A Assembly
DL8593625	Cartridge, Impulse, M29A2 Assembly

U.S. Air Force

SPI 01-364-3955	Remover, Cartridge Actuated RAU-3/A

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, 101 Strauss Avenue, Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to the first article inspection (see 4.4 and 6.3).

3.2 Materials. All materials used in the manufacture of the removers and release mechanisms shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

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3.2.1 Raw material. The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

3.3 Hardness and other physical properties. Hardness and other physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.4 Components.

3.4.1 Primary components. For the purpose of this specification the M29A2 Impulse Cartridge (Drawing 8593625) and the M1A1 Release (Drawing 5184966) are defined as primary components. Only primary components from a single lot shall be used in a production lot of removers; however, one primary component production lot may be used in more than one remover production lot.

3.4.2 Component parts. The M1A1 Release shall meet the requirements of MIL-R-82689 and the following:

3.4.2.1 Hydrostatic pressure. The components listed in table I shall withstand a constant internal pressure, as indicated, for 15 seconds minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.5.1).

TABLE I. Hydrostatic pressure requirements.

Component	Hydrostatic pressure (psi, minimum)
Outside tube	2900
Telescoping tube	2000
Inside tube	2500

3.4.2.2 Torque. Torque requirements shall be as specified in tables II and III. No loosening or deformation shall occur upon application of the breakaway torque.

3.4.2.2.1 Assembly torque. Torque requirements shall be as specified in table II. No movement shall occur upon application of the breakaway torque.

TABLE II. Torque requirements for assembly.

Components	A	B	C
	Run-ON, min (inch pounds)	Seating or bottoming, including A (inch pounds)	Breakaway, min (inch pounds)
Firing head and inside tube with cartridge installed	10	400 + 50	300

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3.4.2.2.2 Component torque. Torque requirements shall be as specified in table III. No loosening or deformation shall occur upon application of the breakaway torque.

TABLE III. Torque requirements for components.

Components	A	B	C
	Run-ON, min (inch pounds)	Seating or bottoming, including A (inch pounds)	Breakaway, min (inch pounds)
Base and outside tube	75 min	500 + 50	400 min
Firing head and inside tube	75 min	—	—
Stop telescoping tube and outside tube	—	—	300 min

3.4.2.3 Assembly. The boss face surface of the firing head shall be in complete contact with the cartridge.

3.4.2.4 M1A1 firing pin release. Each M1A1 used in the RAU-3/A remover shall have met the inspection and test requirements of MIL-R-82689 and DL5184966 prior to installation and shall have been assembled within the past 6 months.

3.5 Protective finish. Protective finish shall be in accordance with the applicable specifications and shall apply to all surfaces of the component regardless of contour configuration and includes interior surfaces. Special internal electrodes may be required to get the proper internal coating.

3.6 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction (see 4.5.3.5).

3.7 Radiographic examination. Each remover shall be radiographically examined for defects (see 4.6.2).

3.8 Functional.

3.8.1 Ballistics. Removers loaded with accepted M29A2 cartridges shall propel a 300-pound mass horizontally and shall comply with the requirements of table IV.

3.8.2 Initiation. The remover shall be fired by a source delivering an output pressure of 250 + 25, − 0 pounds per square inch gage (psig) at the end of 15 feet of aircraft hose, MS28741-4 at 70 ± 5°F.

3.8.3 Misfires. There shall be no misfires.

3.8.4 Mechanical failures. The remover shall function without mechanical failure (see 6.5.1).

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TABLE IV. Ballistic requirements.

Temperature (°F)	Velocity, min (fps)	Thrust, min (pounds)	Ignition delay, max (milliseconds)
− 60 to − 70	17	1930	40
65 to 75	20	2745	40
195 to 205	24	3345	40

3.9 Workmanship. Removers shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.9.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.9.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.9.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)
c. Remanufacture program inspections (see 4.6)

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: room ambient (68 to 77°F)
b. Altitude: sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psi

4.4 First article inspection. At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of five metal parts assemblies, nine sets of unassembled parts, and one set of unassembled parts without plating or final protective finish. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the first article sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new first article samples until an acceptable sample is submitted (see 6.3).

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.5.1.2 Size of lot. The production lot size shall be as specified in the contract.

4.5.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

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4.5.2.1 Cartridge, Aircraft Canopy Remover, M29A2, Drawing 8593625, covering a detail of Drawing 6783541.

Characteristics	Defect	Method of Inspection
Critical		
1	Mixed cartridges	Visual
2	Metal defective	Visual
3	O-ring missing or improperly secured	Visual
4	Sealing disc missing or damaged	Visual
5	Marking missing, incorrect, or illegible	Visual
Major		
101	Cartridge discolored, dirty, oily, corroded, or stained	Visual
102	Case dented	Visual
103	O-ring split, cut, or cracked	Visual
Minor		
201	Case scratched, wrinkled, or bulged	Visual
202	Varnish missing or inadequate	Visual

4.5.2.2 Assembly, Drawing 6783541, less all drawings except 8593729, 6783601, 6783542, 8593794, and AN 565DC1032H2.

Critical		
None defined		
Major		
101	Thread damaged	Visual
102	Burr	Visual
103	Foreign matter	Visual
Minor		
None defined		

4.5.2.3 Assembly, Drawing 6783541, less Drawings 8593658, 8593734, 8593786, 8593787, and 8593791.

Critical		
1	Boss face surface of firing head not in complete contact with cartridge	Gage
2	Cartridge missing	Visual
3	Seal on inside tube missing	Visual

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Major		
101	Burr	Visual
102	Foreign matter	Visual
Minor		
None defined		

4.5.2.4 Assembly, Drawing 6783541, less Drawing 5184966.

Critical		
1	Tubes not free to slide	Manual
Major		
101	Concentricity of body, large end, with datum location	Gage
Minor		
None defined		

4.5.2.5 Assembly, Drawing 6783541.

Critical		
1	Improper assembly	Visual/X-ray
2	Any part missing	Visual/X-ray
3	Mixed removers	Visual
4	M1A1 firing pin release not installed properly	Visual
5	Marking missing, incorrect, or illegible	Visual
Major		
101	Total length, max	Gage
102	Locking wire missing or improperly secured	Visual
103	Safety pin missing	Visual
Minor		
201	Location of labels incorrect	Visual
202	Protective finish inadequate or defective	Visual

4.5.2.6 Seal, Plain, Drawing 8593658, covering a detail of Drawing 6783541.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Diameter of wire, max	Measure
102	Outside diameter, min	Measure

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103	Location of variance of ends, max	Visual
104	Number of coils incorrect	Visual
105	Metal defective	Visual
106	Protective finish inadequate or defective	Visual
107	Burr, sharp edge	Visual
108	Foreign matter	Visual
Minor		
201	Outside diameter, max	Measure
202	Length between coils, incorrect	Measure
203	Direction of turn incorrect	Visual
204	Radius or chamfer missing or incorrect	Visual

4.5.2.7 Spring, Firing Pin, Drawing 8593729, covering a detail of Drawing 6783541.

Critical
None defined

Major		
101	Load at specified heights	Gage
102	Outside diameter	Gage
103	Ends incorrect	Measure
104	Metal defective	Visual
105	Protective finish inadequate or defective	Visual
106	Burr, sharp edges, and slivers	Visual
107	Foreign matter	Visual
Minor		
201	Closed length, max	Gage
202	Diameter of end coils, both ends	Measure

4.5.2.8 Tube, Telescoping, Drawing 8593734, covering a detail of Drawing 6783541.

NOTE
Examinations and tests marked with an asterisk are the minimum requirements for remanufactured RAU-3/A Removers.

	Critical		
*	1	Metal defective	Visual
	Major		
	101	Concentricity of body, large end, with datum location	Gage
	102	Diameter of body, large end, min	Gage
	103	Diameter of body, small end, min	Gage
	104	Diameter, inside, max	Gage
	105	Depth of spiral groove	Gage

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	106	Depth of spiral groove relief	Gage
	107	Total length, max	Gage
	108	Variation in wall thickness	Gage
	109	Straightness	Gage
	110	Metal defective	Visual
*	111	Protective finish inadequate or defective	Visual
*	112	Burr	Visual
*	113	Foreign matter	Visual
	Minor		
	201	Diameter of body, large end, max	Gage
	202	Diameter of body, small end, max	Gage
	203	Diameter, inside, min	Gage
	204	Total length, min	Gage
	205	Length of spiral groove relief, max	Measure
	206	Length of body, large end	Measure
	207	Width of spiral groove relief, max	Measure
	208	Location of spiral groove	Measure
	209	Number of turns of spiral groove incorrect	Visual
	210	Direction of turn of spiral groove incorrect	Visual
	211	Profile of spiral groove incorrect	Visual
	212	Marking missing, incorrect, or illegible	Visual
	213	Radius missing or incorrect	Visual
	214	Surface finish improper	Visual

4.5.2.9 Stop, Telescoping Tube, Drawing 8593786, covering a detail of Drawing 6783541.

Critical
None defined

	Major		
*	101	Pitch diameter of large thread, max	Gage
*	102	Minor diameter of large thread, max	Gage
*	103	Pitch diameter of small thread, max	Gage
*	104	Minor diameter of small thread, max	Gage
	105	Concentricity of bore with pitch diameter of large thread	Gage
	106	Concentricity of counterbore with bore	Gage
	107	Concentricity of hole on body with hole at flat	Gage
	108	True position of threaded holes with hole at flat	Gage
	109	True position of hole at flat with bore	Gage
	110	Pitch diameter of setscrew hole	Gage
	111	Location of setscrew hole	Gage
	112	Diameter of thread undercut, max	Gage
	113	Diameter of hole on flat	Gage
	114	Diameter of hole in body	Gage
	115	Diameter of body, min	Gage

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	116	Diameter of bore, max	Gage
	117	Diameter of counterbore, max	Gage
	118	Depth of thread cavity	Gage
	119	Depth of counterbore, max	Gage
	120	Location of hole on flat	Gage
	121	Location of flat, min	Gage
*	122	Thread damaged or not full	Visual
*	123	Metal defective	Visual
*	124	Protective finish inadequate or defective	Visual
*	125	Burr or sharp corner	Visual
*	126	Foreign matter	Visual
	Minor		
	201	Pitch diameter of large thread, min	Gage
	202	Pitch diameter of small thread, min	Gage
	203	Diameter of bore, min	Gage
	204	Diameter of body, max	Gage
	205	Diameter of counterbore, min	Gage
	206	Diameter of car seal hole	Gage
	207	Depth of flat	Gage
	208	Depth of counterbore, min	Gage
	209	Total length	Gage
	210	Location of flat, max	Gage
	211	Angular location of car seal hole	Gage
	212	Width of thread undercut, max	Measure
	213	Marking missing, incorrect, or illegible	Visual
	214	Radius or chamfer missing or incorrect	Visual
	215	Surface finish improper, except as otherwise classified	Visual

4.5.2.10 Base, Drawing 8593787, covering a detail of Drawing 6783541.

Critical
None defined

	Major		
*	101	Pitch diameter of thread, max	Gage
*	102	Minor diameter of thread, max	Gage
	103	Concentricity of body with pitch diameter of thread	Gage
	104	True position of holes with pitch diameter of thread	Gage
*	105	Diameter of thread undercut, max	Gage
	106	Diameter of body, min	Gage
	107	Thickness of base, min	Gage
*	108	Thread damaged or not full	Visual
*	109	Metal defective	Visual
*	110	Protective finish inadequate or defective	Visual
*	111	Burr	Visual

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*	112	Foreign matter	Visual
	Minor		
	201	Pitch diameter of thread, min	Gage
	202	Diameter of body, max	Gage
	203	Diameter of holes	Gage
	204	Length of body	Gage
	205	Thickness of base, max	Gage
	206	Diameter of flange	Measure
	207	Total length	Measure
	208	Width of thread undercut, max	Measure
	209	Marking missing, incorrect, or illegible	Visual
	210	Radius or chamfer missing or incorrect	Visual
	211	Surface finish improper	Visual
	212	Foreign matter	Visual

4.5.2.11 Tube, Inside, Drawing 8593788 and 8593789, covering a detail of Drawing 6783541.

Critical
None defined

	Major		
*	101	Pitch diameter of thread, min	Gage
*	102	Major diameter of thread, min	Gage
	103	Concentricity of pitch diameter of thread with inside diameter at datum location	Gage
	104	Diameter, inside, max	Gage
	105	Diameter of body, large end	Gage
	106	Diameter of body midsection, min	Gage
	107	Depth of spiral groove	Gage
	108	Depth of spiral groove relief	Gage
	109	Total length, max	Gage
	110	Length of effective thread, min	Gage
	111	Thickness of wall, min	Gage
	112	Thickness of wall under pellet, min	Gage
	113	Straightness	Gage
*	114	Pellet missing, loose, or improperly inserted	Visual
*	115	Thread damaged or not full	Visual
	116	Metal defective	Visual
*	117	Protective finish inadequate or defective	Visual
	118	Surface finish of body, large end, improper	Visual
*	119	Burr	Visual
*	120	Foreign matter	Visual

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	Minor		
	201	Pitch diameter of thread, max	Gage
	202	Diameter, inside, min	Gage
	203	Diameter of body, midsection, max	Gage
	204	Total length, min	Measure
	205	Length of shoulder	Measure
	206	Length of spiral groove relief, max	Measure
	207	Length of thread, max	Measure
	208	Width of spiral groove relief, max	Measure
	209	Location of spiral groove, max	Measure
	210	Profile of spiral groove	Visual
	211	Direction of turn of spiral groove incorrect	Visual
	212	Number of turns of spiral groove incorrect	Visual
	213	Marking missing, incorrect, or illegible	Visual
	214	Radius or chamfer missing or incorrect	Visual
	215	Surface finish improper, except as otherwise classified	Visual

4.5.2.12 Tube, Outside, Drawing 8593790 and 8593791, covering a detail of Drawing 6783541.

Critical
None defined

	Major		
*	101	Pitch diameter of thread, min (both ends)	Gage
*	102	Major diameter of thread, min (both ends)	Gage
	103	Concentricity of pitch diameter of thread with inside diameter at datum location (both ends)	Gage
	104	Diameter, inside, max	Gage
	105	Diameter of body, min	Gage
	106	Total length, min	Gage
	107	Thickness of wall, min	Gage
	108	Length of effective thread, min (both ends)	Measure
	109	Straightness	Gage
*	110	Metal defective	Visual
*	111	Protective finish inadequate or defective	Visual
*	112	Burr	Visual
*	113	Foreign matter	Visual
*	114	Pellet missing, loose, or improperly installed	Visual
	Minor		
	201	Pitch diameter of thread, max	Gage
	202	Diameter, inside, min	Gage
	203	Diameter of body, max	Gage
	204	Total length, max	Gage
	205	Length of thread, max (both ends)	Measure
	206	Marking missing, incorrect, or illegible	Visual

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	207	Radius or chamfer missing or incorrect	Visual
	208	Surface finish improper	Visual

4.5.2.13 Head, Firing, Drawing 6783601, covering a detail of Drawing 6783541.

	Critical		
	1	Metal defective	Visual
	Major		
*	101	Pitch diameter of thread, max	Gage
*	102	Minor diameter of thread, max	Gage
	103	Concentricity of cavity with pitch diameter of thread	Gage
	104	Concentricity of body with pitch diameter of thread	Gage
	105	Concentricity of flange with body	Gage
	106	True position of safety pin hole with cavity including perpendicularity with hole in body	Gage
	107	True position of hole in body in cavity	Gage
	108	True position of flange holes with body, small end	Gage
	109	Diameter of cavity	Gage
	110	Diameter of safety pin hole	Gage
	111	Diameter of flange, min	Gage
	112	Diameter of hole in body	Gage
	113	Diameter of body, min	Gage
	114	Thickness of flange, min	Gage
	115	Thickness of wall, min	Gage
	116	Location of hole in body	Gage
	117	Location of safety pin hole	Gage
	118	Diameter of body, small end, min	Measure
*	119	Thread damaged or not full	Visual
*	120	Protective finish inadequate or defective	Visual
*	121	Surface finish of cavity improper	Visual
*	122	Burr or sharp corner	Visual
*	123	Foreign matter	Visual
	Minor		
	201	Pitch diameter of thread, min	Gage
	202	Diameter of seal hole in flange	Gage
	203	Diameter of flange, max	Gage
	204	Diameter of body, max	Gage
	205	Diameter of thread undercut, max	Gage
	206	Depth to bottom of threaded cavity	Gage
	207	Depth to bottom of long cavity	Gage
	208	Thickness of flange, max	Gage
	209	Height of boss	Gage
	210	Total length	Gage
	211	Length to top of flange	Gage

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212	Location and angle of seal hole in flange	Gage
213	Diameter of body undercut, min	Measure
214	Diameter of body, small end, max	Measure
215	Diameter of boss flat	Measure
216	Diameter of flange holes	Measure
217	Diameter of drill point, max	Measure
218	Depth to drill point, max	Measure
219	Depth of flange holes	Measure
220	Location of flange holes	Measure
221	Angle of boss, incorrect	Measure
222	Width of body undercut, max	Visual
223	Width of thread undercut, max	Visual
224	Marking missing, incorrect, or illegible	Visual
225	Radius or chamfer missing or incorrect	Visual
226	Surface finish improper, except as otherwise classified	Visual

4.5.2.14 Pin, Safety, Drawing 6783542, covering a detail of Drawing 6783541.

Critical
None defined

Major

*	101	Diameter of wire	Measure
*	102	Profile incorrect	Visual
*	103	Metal defective	Visual
*	104	Protective finish inadequate or defective	Visual
*	105	Burr	Visual
*	106	Foreign matter	Visual

Minor

*	201	Diameter of head, inside	Measure
*	202	Total length	Measure
*	203	Length of loop to end of twist	Measure
*	204	Length to center of loop	Measure
*	205	Length between high and low prongs	Measure
*	206	Location of end band	Measure
*	207	Profile of end incorrect	Visual
*	208	Radius missing or incorrect	Visual
*	209	Surface finish improper	Visual

4.5.2.15 Pin, Firing, Drawing 8593794, covering a detail of Drawing 6783541.

Critical
None defined

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	Major		
	101	Concentricity of striker with body	Gage
	102	Concentricity of rear striker with body	Gage
	103	True position of pin hole with body	Gage
	104	True position of cross pin hole with body including perpendicularity with pin hole	Gage
	105	Diameter of body, min	Gage
	106	Diameter of hole	Gage
	107	Length of striker	Gage
	108	Length between holes	Gage
	109	Location of hole	Gage
*	110	Metal defective	Visual
*	111	Protective finish inadequate or defective	Visual
*	112	Surface finish of body improper	Visual
*	113	Burr or sharp corners	Visual
*	114	Foreign matter	Visual
	Minor		
	201	Diameter of striker	Gage
	202	Diameter of body, max	Gage
	203	Diameter of chamfer at hole	Gage
	204	Total length	Gage
	205	Angle of chamfer on holes	Measure
	206	Marking missing, incorrect, or illegible	Visual
	207	Radius or chamfer missing or incorrect	Visual
	208	Surface finish improper, except as otherwise classified	Visual

4.5.2.16 Assembly, Drawing 6783541.

	Critical		
*	1	Firing pin not cocked	Manual/Visual
*	2	Firing pin spring missing	Manual/Visual
	Major		
*	101	Any part missing, except as otherwise classified	Visual
*	102	M1A1 Firing pin release not properly installed	Visual
*	103	Improper assembly	Visual
*	104	Thread damaged or not full	Visual
*	105	Marking missing, incorrect, or illegible	Visual
*	106	Burr	Visual
*	107	Foreign matter	Visual
	Minor		
*	201	Protective finish inadequate or defective	Visual

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* **4.5.3 Inspection testing.**

4.5.3.1 Hardness. Each each heat treated lot of components shall be tested for compliance with the requirements on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.5.3.2 Hydrostatic pressure. Each tube shall be subjected to the hydrostatic pressure test of table I. Failure of any cap or chamber subassembly to comply with 3.4.2.1 shall be cause for rejection of the subassembly.

4.5.3.3 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.5.3.4 Breakaway torque. A breakaway torque shall be applied to each remover threaded joint listed in table II. Failure of any remover to comply with the requirements of 3.4.2.2 shall be cause for rejection.

4.5.3.5 Residual magnetism. Each remover shall be tested for excessive residual magnetism. Failure of any remover to comply with the requirements of 3.6 shall be cause for rejection.

4.5.3.6 Functional. The tests of 4.5.3.6.1 shall be performed by the Government.

4.5.3.6.1 Ballistic. Twenty-one units shall be tested to determine compliance with the requirements of 3.8. The units shall be temperature conditioned as provided in table V. Failure of any unit to comply with the requirements of 3.8 shall be cause for rejection of the lot.

TABLE V. Conditioning temperature.

Temperature (°F)	Number of Samples
− 65 to − 70	7
65 to 75	7
195 to 205	7

4.5.3.6.2 Retest. There shall be no retests.

4.5.3.6.3 Test failure. If a test failure is attributable to an assignable cause, excluding the test remover, the original test results shall be discarded and that test reconducted.

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4.6 Test methods and procedures.

4.6.1 Reused parts. Exceptions as to which parts can be used in remanufacture are as follows:

a. Only threaded tubes that have provisions for the use of Nyloc pellets are acceptable.

b. Nyloc pellets are to be removed before any surface treatments and replaced with new pellets once refurbished.

c. Telescoping tube stops that do not contain a setscrew hole shall be modified to contain the #10-32UNF-3A setscrew hole in accordance with Drawing 8593786.

d. Outside tubes containing nylon pellets at the telescoping tube stop end are acceptable provided the nylon pellets at this end are removed.

4.6.1.1 New parts. The reclaimed parts listed below are not suitable for use in remanufacture and must be replaced with new parts:

a. Seal, Plain, Drawing 8593658
b. Seal, Plain, Drawing 8593659
c. Spring, Firing Pin, Drawing 8593729
d. Pin, Safety, Drawing 6783542
e. Wire, Locking, Drawing 8594748
f. Pellet, Drawing 8594155
g. Setscrew, Black Oxide Coated, AN 565DC1032H2

4.6.1.2 Touch-up. Internal aluminum components with exposed base metal shall be treated with chemical film per MIL-C-5541, Grade A or B, Class 1.

4.6.2 Radiographic inspection. All assembled removers shall be radiographically examined in accordance with MIL-STD-453, with the exception that an assembled RAU-3/A remover shall be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The remover shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the x-ray machine. All removers shall be identified with serial numbers prior to x-ray examination. The serial numbers shall be in consecutive numerical order beginning with the number "001" in each production lot. Any missing serial numbers are to be identified on the x-rays. The removers shall be arranged on boards or trays in consecutive order and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic identification shall include item nomenclature, the complete lot number assigned to that lot, and the span of serial numbers displayed. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance tests. Defective removers found during x-ray review are to be marked on the x-ray prior to shipment and those defective items shall be removed

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from the production lot. Film interpretation of the radiographs shall verify the presence and correct positioning of the following parts:

(1) Pin, Firing	P/N 8593794
(2) Head, Firing	P/N 6783601
(3) Spring, Firing Pin	P/N 8593729
(4) Stop, Telescoping Tube	P/N 8593786
(5) Tube, Inside and Pellet Assembly	P/N 8593789
(6) Tube, Telescoping	P/N 8593734
(7) Seal, Plain	P/N 8593659
(8) Base	P/N 8593787
(9) Seal, Plain	P/N 8593658
(10) Tube, Outside and Pellet Assembly	P/N 8593791
(11) Pin, Safety	P/N 6783542
(12) Setscrew	AN 565DC1032H2
(13) Sealer	P/N 10520495
(14) Wire, Locking	P/N 8594748
(15) Cartridge, Impulse, M29A2	P/N 8593625
(16) Label, Identification	P/N 6783600
(17) Label, Warning	P/N 6783599
(18) M1A1 Firing Pin Release	P/N 5184966

4.6.3 Functional.

4.6.3.1 Ballistics. The test sample shall be rigidly supported so that the entire thrust is borne by the mounting points. The samples shall be tested within 10 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned at the original temperatures.

4.6.3.2 Temperature conditioning. Each assembled remover shall be conditioned for not less than 4 hours nor greater than 24 hours. When reconditioning removers at −65°F, all condensation shall be removed from the exterior surfaces before returning the remover to the conditioning chamber.

4.7 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with SPI 01-364-3955.

5.2 Lot numbers. Lot numbers shall be in accordance with MIL-STD-1168.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

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6.1 Intended use. The remover assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. Production test activity and lot size (see 4.1 and 4.2.1)

d. If first article inspection is required (see 3.1 and 6.3)

e. Directions for shipping radiographs of entire production lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.3.3)

f. Level of packing and marking (see 5.1)

g. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5 are applicable.

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors that the items should be a first article sample. The first article should consist of five metal parts assemblies, nine sets of unassembled parts, and one set of unassembled parts without plating or final protective finish. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Hazard notice. The remover described herein and certain of its components are flammable or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.5 Definition.

6.5.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

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6.5.2 Load firing. The expected performance of the remover, under conditions simulating those existing in the aircraft, is obtained when the remover satisfactorily meets the conditions of the "load" firing.

6.6 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

Custodian:
Navy - OS
Air Force - 99

Preparing activity
Navy - OS
(Project 1377-0E21)

Review activity:
Air Force - 70

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2. 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-82879	931220

3. DOCUMENT TITLE
REMOVER, CARTRIDGE, ACTUATED RAU-3A ASSEMBLY

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Commander, Indian Head Division Naval Surface Warfare Center (Code 8420)	301-743-4358/4510	354-4510
c. ADDRESS (Include Zip Code) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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